

The MLP baseline achieved a test accuracy of 97.79%, while the base CNN reached 99.04%, an improvement of 1.25%. The CNN performs better because it preserves spatial structure through convolutional layers, allowing it to learn local patterns like edges and shapes directly from the image pixels, whereas the MLP just flattens the image and loses spatial relationships between pixels.

Run Name	Architecture	Kernel Size	Filters	Parameter s	Test Accuracy
MLP Baseline	MLP	N/A	N/A	235,146	97.79%
CNN	CNN	3x3	32-64	421,642	99.04%
CNN 5x5	CNN	5x5	32-64	454,922	99.31%
CNN 16_32	CNN	3x3	16-32	206,922	99.16%
CNN 5x5_16_32	CNN	5x5	16-32	215,370	99.09%

In Part 4, the experiments compared a larger CNN (5×5 filters, 32–64 channels) with a smaller CNN (3×3 filters, 16–32 channels). The expectation was that the 5×5, 32–64 model would perform better due to its higher capacity and larger receptive fields, allowing it to capture more spatial context, while the 3×3, 16–32 model would be more lightweight but potentially perform slightly worse. MLflow metrics showed that the larger model achieved slightly higher test accuracy and lower training loss, but the improvement was minute, indicating diminishing returns. In Part 5, the CNN (5×5, 16–32) was tested to isolate the effect of filter size while keeping capacity lower; the expectation was that it would perform between the two Part 4 models. The results were slightly surprising, with MLflow showing accuracy slightly worse than both the 5×5 32–64 model and the 3×3 16-32 model, suggesting that increasing filter size alone without sufficient channel capacity did not improve performance and may have reduced efficiency in learning useful features.

The learned filters in the first convolutional layer resemble simple edge detectors and stroke patterns, highlighting horizontal, vertical, and diagonal lines similar to parts of the handwritten digits seen in the images. Deeper layer filters appear more abstract, combining these low-level features into more complex shapes. The feature maps display how different regions of the input image activate specific filters. Early maps tend to emphasize the edges and contours of the digits, while later maps focus on higher-level structures like loops or intersections, looking at the most significant aspects of the digits for classification.

Something that surprised me was how small the performance differences were across architectures, even when increasing filter size or model capacity. MLflow metrics showed that all CNN variants performed very similarly on MNIST. If given another week, the next steps would be to test more impactful changes such as experimenting with different architecture builds.